Test-Time Score Distillation: Mask-Free Semantic Style Editing with Reverse-Process Regularization Supplementary Material

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A. Overview

This supplement provides the score-contrast derivation, inference settings, design controls, reference-data protocol, and complete numerical records. It accompanies the initial 40-reference study and expanded development comparisons in the main paper. The frozen multi-seed test is being generated, and its final results will use the same pairing and metric definitions.

New references have identifiers prefixed by `kr_dev` and `kr_test`. The initial 40-reference, single-prompt/seed study has a separate section with complete per-reference measurements. That archive contains 160 stored measurements. The original output images and complete run/evaluator records are not all recovered, so the archived table and new experiments have separate provenance.

B. Derivation of the latent objective

B.1. Conditioning and score conversion

Let r be the reference and c the full target instruction. At the same state z_σ, define $v_r = v_\theta(z_\sigma, \sigma, c, r)$ and $v_b = v_\theta(z_\sigma, \sigma, c, \varnothing)$. Removing the reference does not remove the text. Target geometry, latent, and noise level are shared. Both predictions use frozen weights and are detached during latent optimization.

For $z_\sigma = (1-\sigma)x_0 + \sigma\epsilon$, with Gaussian ϵ, the conditional-mean flow velocity obeys

$$v(z_\sigma, \sigma) = \mathbb{E}[\epsilon - x_0 \mid z_\sigma]. \tag{S1}$$

The conditional Gaussian-noise score identity is $s(z_\sigma, \sigma) = -\mathbb{E}[\epsilon \mid z_\sigma]/\sigma$. Taking the conditional mean of $z_\sigma + (1-\sigma)(\epsilon - x_0) = \epsilon$ therefore gives

$$s_\theta = -\frac{z_\sigma + (1-\sigma)v_\theta}{\sigma}. \tag{S2}$$

The learned velocity approximates this conditional-mean field. Subtracting the two conditions cancels the state term:

$$\Delta s = s_r - s_b = -\frac{1-\sigma}{\sigma}(v_r - v_b). \tag{S3}$$

For a positive noise level below one, the velocity-to-score contrast factor is negative. Optimization is omitted at the terminal zero-noise state, and the numerical implementation clamps small denominators.

B.2. Detach, reduction and update magnitude

For each sample with N latent elements, let $g = a\Delta s$, $a > 0$, and

$$d = \frac{g}{\max(\sqrt{N^{-1}\sum_j g_j^2}, \varepsilon_g)}. \tag{S4}$$

The primary branch uses a detached velocity to define

$$\hat{x}_0(z) = z - \sigma\,\mathrm{sg}(v_b), \tag{S5}$$

$$y = \mathrm{sg}(\hat{x}_0(z) + \eta d), \tag{S6}$$

$$L_p = \frac{1}{2N}\sum_j (\hat{x}_{0,j}(z) - y_j)^2. \tag{S7}$$

Both d and y are treated as constants during differentiation. Hence

$$\frac{\partial \hat{x}_0}{\partial z} = I, \qquad \nabla_z L_p = -\frac{\eta}{N} d. \tag{S8}$$

The identity Jacobian follows from detaching v_b in this branch. Without that detach, an additional velocity Jacobian would enter. The factor $1/N$ follows from mean reduction and must not be dropped when interpreting SGD.

Plain SGD with learning rate ℓ and no other losses produces $\Delta z = \ell\eta d/N$. Adam or AdamW instead rescales gradients using moment estimates and numerical stabilization. Thus η fixes a target offset, not an optimizer displacement. Positive multipliers cancel in RMS normalization away from the numerical floor. We record actual update RMS to characterize displacement under the chosen optimizer and precision.

For the evaluated AdamW setting, weight decay is zero. Let $h_k = \nabla_z L_p$ at inner iteration k. Starting from $m_0 = u_0 = 0$ at each outer intervention, the elementwise update

is

$$m_k = \beta_1 m_{k-1} + (1-\beta_1) h_k, \tag{S9}$$

$$u_k = \beta_2 u_{k-1} + (1-\beta_2) h_k^2, \tag{S10}$$

$$\hat{m}_k = m_k/(1-\beta_1^k), \qquad \hat{u}_k = u_k/(1-\beta_2^k), \tag{S11}$$

$$z_{k+1} = z_k - \ell \frac{\hat{m}_k}{\sqrt{\hat{u}_k} + \epsilon_{\text{opt}}}. \tag{S12}$$

The implementation uses the optimizer defaults $\beta_1 = .9$, $\beta_2 = .999$, and $\epsilon_{\text{opt}} = 10^{-8}$. These moments persist within an intervention and are reset at the next outer intervention. The gradient scale and target offset therefore do not specify a fixed latent displacement.

C. Execution order and controls

C.1. Reverse-first alternation

At each actual solver interval, the frozen reverse step first reaches the next noise level. An eligible positive reached node is then assigned its inner latent updates. The teacher predictions are recomputed after each latent change at that same noise level. A new latent optimizer is created for each outer intervention. Its state persists within that intervention. The final reverse interval has no subsequent inner update.

The evaluated operator uses only the pseudo-target objective. Distribution and content-loss weights are zero, and SLT, warm-start AdaIN, and final AdaIN are disabled. Earlier archived settings have separate configuration provenance.

C.2. Separate text guidance from reference guidance

Sampler text CFG forms

$$v^\beta = v_{\text{uncond}} + \beta(v_{\text{cond}} - v_{\text{uncond}}). \tag{S13}$$

Direct reference CFG forms

$$v_{\text{refCFG}} = v_b^\beta + \gamma(v_r^\beta - v_b^\beta). \tag{S14}$$

The native reference control at $\gamma = 1$ evaluates only the reference branch, and its cost reflects those evaluations. The current protocol records sampler text CFG $\beta = 4$, a style-contrast CFG parameter of 4, and embedded guidance 0 as separate settings. The code forms the positive score multiplier as $a = \lambda_{\text{CA}}(4-1) = 3$ with $\lambda_{\text{CA}} = 1$ before RMS normalization.

C.3. Development controls and comparison settings

The same-backbone controls use a frozen Klein-9B-base pipeline at 512×512, with layer-wise CPU offload and a requested 12-GiB model-memory budget. The common-node variants use one inner update at $\mathcal{G} = \{.8, .6, .4, .2\}$, AdamW at learning rate .01, zero weight decay, and target offset .05. Inserting these nodes yields 29 actual intervals for a nominal 25-step solver and 12 for a nominal eight-step solver. The RMS-SGD control normalizes latent-gradient RMS before an SGD step at rate .01. Measured displacement is retained for each optimizer.

Dense variants use the original positive solver nodes. The 25-step variants vary inner count and learning rate at target offset .05. The eight-step/K5 setting uses target offset .12. Dense25/K5 requires 290 DiT forwards and 120 latent backward calls, compared with 146/48 for the retained 25/K2 and 86/35 for 8/K5. All use text CFG 4. Direct-reference controls vary reference scale and reverse-step budget, with actual model calls recorded for every output.

AD uses SD1.5, 50 steps, two inner updates, learning rate .015, text CFG 7.5, AdaIN, and bf16. USO uses the style-only FLUX.1-dev interface with pretrained rank-128 LoRA/projector and SigLIP, 25 steps, guidance 4, FP8 weight storage, and bf16 computation. Its content-image input is empty. The backbone, adapter, precision, and offload configuration accompany each method's results.

StyleShot uses the official SD1.5 pipeline, a pretrained style encoder and image adapter, style scale 1, text CFG 7.5, and float32 inference. Its PNDM scheduler has 51 actual evaluation nodes at a nominal 50-step setting. The negative prompt is "monochrome, lowres, bad anatomy, worst quality, low quality". The random seed initializes both image sampling and the style-encoder patch permutation. We use the same reference-plus-text inputs and seed-64 grid as the other methods.

InstantStyle uses the official SDXL style-only block `up_blocks.0.attentions.1`, with adapter scale 1, text CFG 5, 30 Euler steps, and float16 inference. Its target text appends "masterpiece, best quality, high quality". The official negative prompt is "text, watermark, lowres, low quality, worst quality, deformed, glitch, low contrast, noisy, saturation, blurry". Image projection and attention adapter parameters pass strict state-dictionary loading. Existing SDXL weights are reused through a local directory view, with no weight changes.

C.4. Complete development comparisons

The search contains 400 outputs from 25 configurations on four development references, two prompts, and two seeds (Table S1). The initial coverage grid contains 240 outputs from five candidates on eight development references, six prompts, and seed 64. Adding StyleShot and InstantStyle on the identical 48 cells each gives the 336-output comparison (Table S2). Every configuration covers the same cells within its grid. Exact common cells are reused between stages, so the search and coverage counts overlap. All outputs have the three reported metric readouts.

Single- and two-GPU placements share the same model

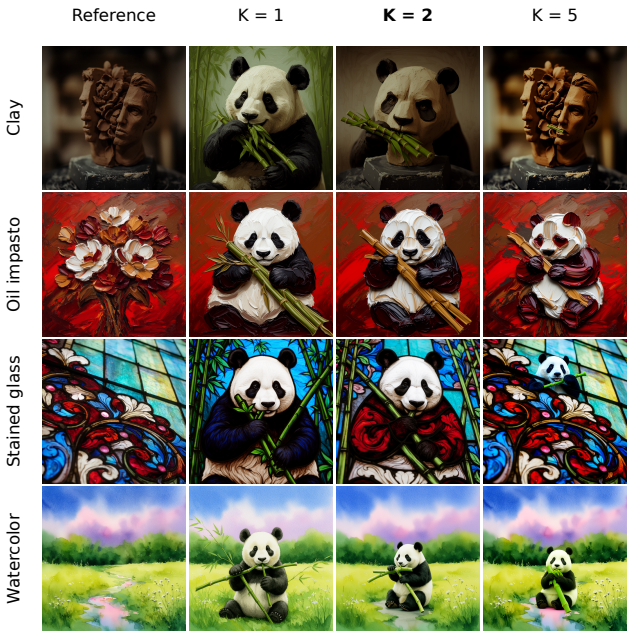


Figure S1. Effect of inner-update count at fixed 25-step sampling, learning rate .01, and target offset .05. Rows cover all four references in the configuration study, using the panda prompt and seed 64. K1, K2, and K5 share the same initial seed and reference. Increasing the update count strengthens the reference appearance, while K2 retains a clearer panda than K5 in the clay example. Table S3 reports means over both prompts and both seeds.

precision and method settings. The two-GPU implementation was checked against a single-GPU reference on a full generation case, with pixel-identical output. Placement and timing are recorded separately.

C.5. Why use a moderate inner-update count?

We vary the inner-update count while keeping the 25-step solver, learning rate .01, target offset .05, backbone, and input cells fixed. K1, K2, and K5 use the same four references, two prompts, and two seeds (16 outputs per setting). K1 gives lower style similarity. Increasing the count to K5 strengthens the reference style but reduces target-text alignment. K2 retains stronger style than K1 with similar CLIP-T and uses fewer forward calls than K5 (Table S3). This comparison motivates the 25/K2 working point used in the formal protocol.

Reducing the K5 learning rate to .005 gives CSD .6288 and CLIP-T .2986, compared with .5555/.3110 for K2 at .01. This control shows that reducing update count and reducing learning rate yield different operating points. The full search table preserves both alternatives. The recorded outputs support paired qualitative inspection of medium appearance and subject structure under these settings.

Figure S1 shows the visual role of update count on the same references. The comparison varies only K, keeping the solver and optimizer settings fixed. It explains the choice of a moderate update budget alongside the complete-grid measurements.

C.6. Frozen held-out evaluation

After the complete development checks, protocol `cvpr_test_v1` freezes 40 disjoint test references, six literal prompts, seeds 64/65/66, and five settings before generating test outputs (720 per setting, 3,600 total). TSD8/K5 uses eight intervals, five updates, target offset .12, and learning rate .01. TSD25/K2 uses 25 intervals, two updates, target offset .05, and learning rate .01. Both use text CFG 4 and disable auxiliary distribution/content losses, SLT, and AdaIN. The direct control uses 72 intervals and reference scale five. AD and USO retain their documented development settings. Sources, input hashes, model-file identities, and the full cell matrix are frozen. No test result is used to tune parameters. The test is running, not complete, in this working supplement.

D. Reference data and metric definitions

D.1. New collection and split

The new Krea collection contains 48 public-feed synthetic references, visually selected for medium clarity, quality, and content diversity before method outputs were examined. Each of eight medium categories contributes one development and five test references (Table S4). Studio/tool photographs, dominant gallery frames, multi-panel collages, and ambiguous media were excluded during reference curation, not according to method success. The category labels describe the visual appearance of these synthetic references.

All methods receive the same 512-by-512 RGB reference, obtained by aspect-preserving Lanczos resizing and center cropping of the public 1024-pixel image version. The manifest records source UUIDs and URLs, source/derived hashes, split, and transform. The numeric and protocol package contains source metadata. It does not redistribute the raw reference images.

D.2. Readouts and their limitations

CSD is cosine similarity between style descriptors of an output and its reference [3]. CLIP-T is image–text cosine similarity to the target content request without the style-instruction wrapper [2]. Reference-layout similarity is Pearson correlation between off-diagonal patch self-similarity vectors from DINO features [1]. Higher layout similarity is task-dependent and may reflect shared semantics as well as geometry. None of these scores alone establishes successful transfer or image quality.

For new evaluation, the implemented protocol specifies the official CSD ViT-L checkpoint, OpenAI CLIP ViT-

Table S1. Complete development search: 25 configurations, each with four references, two prompts, and two seeds (16 outputs). "Four nodes" denotes one update at each common intervention node. "+nodes" denotes node insertion into the nominal solver. Labels map to experiment identifiers in the source records.

Configuration	CSD ↑	CLIP-T ↑	Layout
AD (SD1.5)	0.6789	0.2894	0.3963
USO (FLUX.1, FP8)	0.3378	0.3244	0.3587
TSD25, four nodes	0.0386	0.3190	0.3398
TSD25, K1	0.3726	0.3109	0.3665
TSD, 25/K2	0.5555	0.3110	0.4350
TSD25, K5, $\ell = .005$	0.6288	0.2986	0.4941
TSD8, four nodes	0.0315	0.3168	0.3465
TSD25, K5	0.7302	0.2796	0.5955
TSD, 8/K5	0.3891	0.3110	0.3754
TSD25, four nodes, $\ell = .005$	0.0341	0.3198	0.3394
TSD25, four nodes, $\ell = .02$	0.0468	0.3191	0.3404
Native global	0.0633	0.3100	0.3411
Native semantic	0.0809	0.3070	0.3516
Ref. CFG144, $\gamma = 1$	0.1401	0.3133	0.3524
Ref. CFG72, $\gamma = 3$	0.2545	0.3185	0.3810
Reference CFG, 72/$\gamma = 5$	0.2631	0.3114	0.4137
Ref. CFG12+nodes, $\gamma = .5$	0.0635	0.3145	0.3386
Ref. CFG12+nodes, $\gamma = 1.5$	0.1865	0.3163	0.3616
Ref. CFG25+nodes, $\gamma = .5$	0.0682	0.3163	0.3421
Ref. CFG25+nodes, $\gamma = 1.5$	0.1862	0.3159	0.3587
Ref. CFG25+nodes, $\gamma = 2$	0.2227	0.3170	0.3650
Ref. CFG25+nodes, $\gamma = 3$	0.2517	0.3185	0.3799
Ref. CFG25+nodes, $\gamma = 5$	0.2470	0.3147	0.4092
Ref. CFG25+nodes, $\gamma = 1$	0.1247	0.3150	0.3522
RMS-SGD, four nodes	0.0411	0.3186	0.3397

Table S2. Expanded development comparison: seven settings on eight references, six prompts, and seed 64 (48 outputs per row).

Method	CSD ↑	CLIP-T ↑	Layout
Reference CFG, 72/$\gamma = 5$	0.3190	0.3005	0.3118
AD (SD1.5)	0.7069	0.2692	0.3224
USO (FLUX.1, FP8)	0.4117	0.3178	0.2953
StyleShot (SD1.5)	0.5042	0.2840	0.2703
InstantStyle (SDXL)	0.4947	0.2616	0.2900
TSD, 8/K5	0.4291	0.3078	0.3101
TSD, 25/K2	0.5751	0.2982	0.3503

Table S3. Inner-update count at fixed 25-step settings. Means over the same 16 development cells. Forward calls count DiT evaluations per output.

Updates	CSD ↑	CLIP-T ↑	Forwards
K1	.3726	.3109	98
K2	.5555	.3110	146
K5	.7302	.2796	290

Table S4. New synthetic-reference split, distinct from the archived 40 IDs.

Medium category	Development	Test
clay	1	5
colored_pencil	1	5
mosaic	1	5
oil_impasto	1	5
stained_glass	1	5
paper_cut	1	5
watercolor	1	5
woodcut	1	5
Total	8	40

B/32, and timm DINOv2 ViT-B/14. CSD inputs use bilinear 224-pixel resizing and CLIP normalization. CLIP-T uses the model's standard preprocessing. DINO inputs use bilinear 518-pixel resizing and ImageNet normalization. The evaluator is designed to record checkpoint hashes and require strict state-dict loading. All three metrics executed with validated weights. The earlier archive retains its own evaluator-provenance qualification.

E. Complete archived per-reference measurements

Tables S5–S7 retain every reference ID and all four methods for the three reported readouts. TSD-S/TSD-G map to archived identifiers `q25s`/`pg_n8_s12_r`. Native-S/Native-G map to `kleinsys_sem`/`kleinsys_gst`. Values are rounded only for display. The numeric export retains the stored precision.

F. Pairing, uncertainty and reporting

The reference ID defines the grouping unit for comparisons. In the historical table, each of 40 references contributes four method outputs at one prompt and seed. Means and sample SD describe cross-reference variation under that setting.

For layout-constrained comparisons, each TSD output must satisfy $\Lambda(x_i^{\mathrm{TSD}}, r_i) \leq \Lambda(x_i^{\mathrm{native}}, r_i)$ against its own paired baseline. Only 1/40 semantic and 3/40 global pairs meet this constraint. Subset style differences are reported with their coverage and paired subject readout, alongside the complete reference population.

For new multi-prompt/multi-seed analysis, the implementation resamples reference groups with their repeated prompt/seed observations together. It uses equal-reference mean differences, 10,000 bootstrap draws, and a fixed bootstrap RNG seed 20261106. Intervals are reported with the number of reference groups. Missing generation or evaluation cells and unpaired observations retain their denominators, and scores are not imputed.

G. Cost and reproducibility

End-to-end timing includes reference/text encoding, latent optimization, sampling, and output decoding. Final PNG writing is excluded. Factory loading and lazy first-call model loading have separate measurements, and cold calls are identified explicitly. We summarize actual solver intervals, model forward calls, latent backward calls, peak allocated GPU memory, resolution, precision, offload policy, and physical-device placement. Single- and two-GPU timings are grouped separately. Recorded shared-machine conditions accompany the measurements.

The source-record package contains the full historical metric table, reference metadata, the 400-output search, and the 336-output expanded development grid. The original five-setting coverage contains 240 outputs. The search and coverage sets overlap through exact cell reuse. Exported records preserve reported metrics and input identifiers, with portable path placeholders and hashes linking them to the original files. The formal protocol also records source hashes and model-file identities. Shared pretrained weights remain unchanged throughout inference.

H. Initial prompt extension and design records

The initial submission also contains stored prompt-level summaries, parameter observations, and a residual diagnostic. We retain these records separately from the newly generated development and held-out studies. Their stored values are re-tabulated here, with the available unit of observation specified for each study.

H.1. Five-reference prompt extension

The extension uses five references and four additional content prompts beyond the panda anchor. Table S8 preserves the stored means of CSD and CLIP-T for each mode. All prompt-level means show higher CSD for TSD than their corresponding native editor. The source includes aggregate win counts, but the per-cell prompt-extension outputs and measurement records are not fully recovered. We therefore use this table as the initial prompt-level summary, without recomputing paired uncertainty from aggregate means. Its panda-anchor native CSD means are .3563 and .2869 on five references. The full 40-reference native means are .2672 and .2474, so the two reference populations are reported separately.

H.2. Initial parameter records

Table S9 retains the available parameter-study measurements. The source contains 213 rows across 36 recorded configuration IDs. Most configurations have five rows, while `q25s` has 38. The available counts are displayed explicitly. The record overlaps the initial benchmark and is not a separate set of independent test samples. The new 400-output configuration study provides the matched-input design comparison used in the main paper.

H.3. Reference-contrast residual diagnostic

The initial SRSD diagnostic measures the RMS magnitude of the native reference-conditioned velocity contrast on a generated image. Its stored 20 rows cover four configurations and five references. Recomputing the association with CSD gives Pearson $r = 0.78$ and Spearman $\rho = 0.88$. A larger residual accompanies higher CSD in these records, so a smaller residual does not act as a monotonic style-distance readout for this diagnostic. We retain external CSD, CLIP-T, and layout measurements for the main evaluation. The archived diagnostic is separate from the new evaluator runs.

Table S5. All archived CSD measurements, one prompt and seed 65.

Style	TSD-S	Native-S	TSD-G	Native-G	Style	TSD-S	Native-S	TSD-G	Native-G
00	0.698	0.507	0.700	0.519	20	0.657	0.344	0.546	0.312
01	0.759	0.482	0.731	0.433	21	0.703	0.110	0.547	0.101
02	0.668	-0.020	0.532	0.010	22	0.720	0.484	0.756	0.478
03	0.819	0.283	0.699	0.239	23	0.612	0.268	0.603	0.141
04	0.813	0.077	0.634	0.003	24	0.758	0.083	0.747	0.085
05	0.770	0.462	0.596	0.463	25	0.603	0.423	0.529	0.434
06	0.659	0.043	0.529	0.079	26	0.622	0.356	0.554	0.338
07	0.527	0.211	0.475	0.246	27	0.757	0.259	0.593	0.183
08	0.702	0.007	0.591	0.028	28	0.726	0.385	0.696	0.291
09	0.795	0.415	0.692	0.144	29	0.668	0.328	0.559	0.203
10	0.683	0.166	0.437	0.182	30	0.590	0.216	0.494	0.235
11	0.784	0.400	0.696	0.403	31	0.700	0.102	0.611	0.109
12	0.838	0.518	0.767	0.279	32	0.886	0.461	0.788	0.528
13	0.635	0.134	0.620	0.172	33	0.721	0.284	0.589	0.280
14	0.804	0.327	0.672	0.303	34	0.729	0.426	0.721	0.448
15	0.802	0.148	0.711	0.101	35	0.712	0.278	0.605	0.278
16	0.774	-0.006	0.473	0.023	36	0.781	0.573	0.657	0.601
17	0.735	0.159	0.683	0.144	37	0.655	0.321	0.596	0.314
18	0.764	-0.012	0.505	0.048	38	0.839	0.139	0.718	0.259
19	0.671	0.317	0.469	0.289	39	0.629	0.227	0.467	0.175

Table S6. All archived CLIP-T measurements, one prompt and seed 65.

Style	TSD-S	Native-S	TSD-G	Native-G	Style	TSD-S	Native-S	TSD-G	Native-G
00	0.309	0.319	0.308	0.313	20	0.308	0.317	0.317	0.331
01	0.314	0.300	0.289	0.323	21	0.302	0.317	0.311	0.312
02	0.299	0.305	0.293	0.305	22	0.288	0.332	0.310	0.322
03	0.326	0.313	0.271	0.314	23	0.277	0.346	0.290	0.314
04	0.253	0.324	0.314	0.305	24	0.306	0.322	0.314	0.316
05	0.283	0.331	0.284	0.343	25	0.325	0.328	0.322	0.331
06	0.345	0.308	0.336	0.302	26	0.320	0.337	0.343	0.344
07	0.324	0.311	0.308	0.316	27	0.261	0.310	0.293	0.312
08	0.312	0.329	0.306	0.317	28	0.345	0.338	0.317	0.308
09	0.230	0.296	0.224	0.312	29	0.306	0.331	0.347	0.318
10	0.327	0.311	0.312	0.309	30	0.321	0.317	0.328	0.315
11	0.217	0.318	0.299	0.305	31	0.320	0.322	0.301	0.323
12	0.300	0.326	0.296	0.325	32	0.217	0.309	0.209	0.309
13	0.324	0.304	0.316	0.309	33	0.301	0.319	0.300	0.299
14	0.277	0.330	0.282	0.305	34	0.281	0.304	0.284	0.296
15	0.269	0.314	0.296	0.306	35	0.272	0.289	0.293	0.308
16	0.302	0.331	0.332	0.309	36	0.274	0.326	0.323	0.334
17	0.309	0.314	0.308	0.304	37	0.310	0.313	0.331	0.331
18	0.283	0.327	0.301	0.326	38	0.269	0.328	0.335	0.317
19	0.308	0.331	0.304	0.313	39	0.303	0.306	0.302	0.314

Table S7. All archived reference-layout similarity measurements. Desirability depends on the requested task. One prompt and seed 65.

Style	TSD-S	Native-S	TSD-G	Native-G	Style	TSD-S	Native-S	TSD-G	Native-G
00	0.448	0.385	0.341	0.376	20	0.345	0.355	0.370	0.322
01	0.735	0.346	0.672	0.419	21	0.449	0.351	0.365	0.324
02	0.359	0.303	0.327	0.281	22	0.533	0.260	0.318	0.287
03	0.697	0.291	0.571	0.220	23	0.373	0.265	0.288	0.260
04	0.590	0.333	0.430	0.364	24	0.388	0.246	0.310	0.244
05	0.803	0.327	0.542	0.393	25	0.474	0.374	0.409	0.348
06	0.620	0.372	0.512	0.363	26	0.526	0.342	0.392	0.338
07	0.482	0.380	0.433	0.368	27	0.546	0.330	0.311	0.336
08	0.279	0.231	0.261	0.202	28	0.430	0.273	0.317	0.208
09	0.771	0.597	0.667	0.337	29	0.706	0.274	0.465	0.275
10	0.723	0.483	0.464	0.483	30	0.342	0.327	0.336	0.322
11	0.743	0.286	0.365	0.303	31	0.592	0.240	0.558	0.183
12	0.671	0.433	0.575	0.219	32	0.794	0.340	0.456	0.284
13	0.627	0.342	0.394	0.358	33	0.512	0.280	0.316	0.282
14	0.741	0.242	0.282	0.261	34	0.782	0.337	0.504	0.331
15	0.763	0.237	0.504	0.272	35	0.485	0.266	0.342	0.250
16	0.669	0.250	0.419	0.263	36	0.627	0.356	0.458	0.348
17	0.640	0.287	0.554	0.279	37	0.474	0.300	0.403	0.263
18	0.669	0.246	0.433	0.269	38	0.601	0.270	0.367	0.270
19	0.443	0.292	0.316	0.275	39	0.416	0.230	0.216	0.191

Table S8. Stored five-reference prompt-extension means from the initial study. Each row averages five references. These are prompt-level summaries, separate from the new six-prompt development study.

Prompt	Mode	TSD CSD	Native CSD	TSD CLIP-T	Native CLIP-T
Panda	Semantic	0.7034	0.3563	0.2849	0.3152
Panda	Global	0.6306	0.2869	0.2837	0.3086
Castle	Semantic	0.6294	0.3335	0.2718	0.3050
Castle	Global	0.5575	0.2837	0.2599	0.3126
Cat	Semantic	0.7245	0.3599	0.2284	0.2929
Cat	Global	0.5862	0.3562	0.2573	0.2881
Sunflowers	Semantic	0.6167	0.2676	0.2819	0.3212
Sunflowers	Global	0.4837	0.2577	0.3018	0.3216
Fisherman	Semantic	0.8115	0.5223	0.2328	0.2769
Fisherman	Global	0.7455	0.3394	0.2193	0.3031

Table S9. Available initial parameter-study records, grouped by their original configuration IDs. n counts stored observations. Configurations with different n are not treated as matched-input comparisons.

Archived configuration ID	Available n	CSD ↑	CLIP-T ↑	Layout
s06	5	0.3730	0.3239	0.4047
s08	5	0.5318	0.2900	0.4419
s10	5	0.5300	0.3085	0.3721
pg_n6_s08_r	5	0.4023	0.3200	0.3934
pg_n6_s12_r	5	0.4355	0.3199	0.3831
pg_n8_s08_r	5	0.5615	0.2899	0.4507
pg_n8_s12_r	5	0.6306	0.2837	0.4527
pg_n10_s08_r	5	0.5713	0.2943	0.4201
pg_n10_s12_r	5	0.5919	0.2944	0.4261
pg_n6_s05_rt	5	0.3799	0.3229	0.4089
pg_n6_s08_rt	5	0.3969	0.3235	0.3859
pg_n6_s12_rt	5	0.4375	0.3133	0.3807
pg_n8_s05_rt	5	0.5252	0.2983	0.4370
pg_n8_s08_rt	5	0.5874	0.2857	0.4534
pg_n8_s12_rt	5	0.6446	0.2835	0.4588
pg_n10_s05_rt	5	0.5250	0.3120	0.3666
pg_n10_s08_rt	5	0.5720	0.2979	0.4129
pg_n10_s12_rt	5	0.6100	0.2899	0.4194
q25s	38	0.7146	0.2971	0.5667
se_i3_s03_r	5	0.6390	0.2807	0.4558
se_i3_s05_r	5	0.6970	0.2774	0.5164
se_i3_s08_r	5	0.7012	0.2870	0.5796
se_i5_s03_r	5	0.6947	0.2852	0.5524
se_i5_s08_r	5	0.7388	0.2802	0.6782
se_i8_s03_r	5	0.7117	0.2840	0.6255
se_i8_s05_r	5	0.7357	0.2765	0.6870
se_i8_s08_r	5	0.7607	0.2598	0.7241
se_i3_s03_rt	5	0.6386	0.2773	0.4475
se_i3_s05_rt	5	0.7115	0.2655	0.5262
se_i3_s08_rt	5	0.7152	0.2781	0.5856
se_i5_s03_rt	5	0.7079	0.2682	0.5585
se_i5_s05_rt	5	0.7353	0.2626	0.6359
se_i5_s08_rt	5	0.7873	0.2243	0.7746
se_i8_s03_rt	5	0.7376	0.2623	0.6406
se_i8_s05_rt	5	0.7949	0.2206	0.7872
se_i8_s08_rt	5	0.8227	0.2062	0.8233

I. Additional validation slots

The following slots reserve the remaining analyses in this working draft. They are separate from the completed tables above.

I.1. Held-out baselines and paired analysis

Six-baseline test (pending). Populate the common 720-cell grid for reference CFG, AD, USO, StyleShot, InstantStyle, CSGO, and the two TSD settings. Validate the three added baselines on development data before freezing their extension protocol. Report actual coverage, paired metric differences, reference-group intervals, and results by medium.

Six-baseline qualitative comparison (pending). Add paired grids with a common reference, prompt, and seed for every displayed method. Keep full-image content intact and use the same framing and display size. Additional images should explain medium transfer and the two task goals.

I.2. Direction, normalization, and optimizer

The current record already contains RMS-SGD, AdamW, common-node sampling, and the fixed-solver K study. The remaining controls target the specific role of the score-direction objective.

Directional target controls (pending). Compare the detached directional target with a contrast-norm objective and an unnormalized direction. Match the frozen backbone, inputs, noise nodes, and intervention count, and record actual displacement and model calls. Test positive-scale cancellation separately as an algebraic implementation check.

I.3. Reverse-process regularization

Regularization controls (pending). Evaluate a separately versioned auxiliary-score variant against the frozen-reverse TSD procedure, with matched input cases and recorded auxiliary training and inference costs. Keep each compared latent consistent with its queried noise level. Report style, subject, layout, and visual behavior without assuming equivalence to distribution matching.

I.4. Style locality and reference structure

Locality controls (pending). Extend the completed common-node controls by fixing intervention locations and counts while varying solver resolution. Study intervention density and target offset separately. Report layout together with style and subject alignment. Organize paired examples by the design choice they explain.

Reference representation (pending). If retained in the final analysis, rerun the original reference-position perturbation with an explicit token/position mapping and matched unperturbed control. Group references by predeclared appearance and object-layout characteristics. This slot tests how native reference information enters the operator.

I.5. Independent human judgments

The study design fixes one prompt/seed case for each of the 40 test references before inspecting outcomes, balancing the six prompts and three seeds. TSD25/K2 is paired with six baselines, giving 240 planned image pairs. Each pair receives three judgments from different participants. Anonymous A/B candidates are shown with the reference and target text. Participants answer separate style, subject, and overall-preference questions, with tie and unable-to-judge choices.

Human study (pending). Complete collection before inserting any preference values. Report actual recruitment and valid response counts, all choice counts, and uncertainty that accounts for repeated references and participants. The planned 720 judgments are assignments, not collected observations.

I.6. Perceptual and editor audits

The earlier manuscript also contained VLM judgments and a small public-editor audit. These are distinct from dedicated transfer baselines and independent human judgments.

Perceptual audit reconciliation (pending). Recover and verify the original prompts, checkpoint/model identities, randomized candidate order, raw judge responses, and output hashes before reusing any VLM summary. If refreshed, keep reference-specificity controls and cross-judge agreement separate from human preference.

External-editor audit (pending). If included, restore a matched reference-plus-text comparison with the original public editors only after verifying model versions, execution settings, and source images. Keep this small auxiliary audit separate from the six-baseline test.

I.7. Source reconciliation for initial studies

Remaining initial-study sources (pending). Recover the prompt-extension cell-level outputs and the original parameter-run configurations if those results are used beyond the stored summaries in Sec. H. Preserve the existing CSV values and keep new reruns under a distinct protocol.

I.8. Cost and configuration records

Timing and memory table (pending). Add warm-call latency and peak allocated memory on common hardware, with resolution, precision, offload, actual solver/model calls, and placement. Keep cold loading separate and report timing variability and the number of repeated measurements.

LLM Usage Statement

OpenAI Codex assisted drafting, editing, mathematical and code checks, protocol and experiment-script development, reference curation, qualitative inspection, experiment management, and figure/statistical-analysis code. New references are Krea-generated. Numerical tables are computed from recorded executions. The authors are responsible for verifying the text, references, data provenance, and computations.

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